

The Auditory Sensitivity of the Lemur

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ABSTRACT A single-lever shock avoidance conditioning technique was used to generate auditory behavioral thresholds for *Lemur catta* over the frequency range 100 Hz to 75 kHz. The results indicate a sensitivity of at least 60 dB below one dyne/cm² from 1 kHz to 32 kHz and, throughout the entire frequency range, appear to be compatible with other available prosimian data. A contrast between anthropoids and prosimians indicates a loss in high frequency sensitivity for the former group, but a greater sensitivity for lower frequency stimuli. Analysis of lemur vocalizations indicates a preponderance of relatively low frequency components. Descriptions of the conditioning technique, threshold criteria, and procedures of sound measurement are included.

As part of a long-term investigation of the hearing of prosimians, we have assessed the auditory capabilities of several species of lemurs by means of conditioning techniques. We chose this representative of the suborder prosimii since no quantitative behavioral data on hearing were available for this animal (Vernon, '67) and the Oregon Regional Primate Research Center maintains one of the largest captive populations of lemurs in the world.

In a previous paper (Mitchell, Gillette, Vernon and Herman, '70), we reported on the auditory sensitivity of *Lemur catta*, *L. fulvus*, and *L. macaco* over the frequency range of 100 hertz to 40 kilohertz, using a two-lever positive reinforcement technique patterned after that used by Gourevitch and Hack ('66). The derived data were obtained only after long and exasperating training and testing sessions; we feel this technique is too difficult for the lemur. Further, it did not permit an appropriate degree of stimulus control, resulting in heightened variability and lessened confidence in the data. Accordingly, we adopted a different behavioral technique, similar to the single-lever shock avoidance paradigm reported by Clack and Herman ('63), and replicated these threshold determinations, this time considering only the species *L. catta*. These data (Mitchell, Vernon and Herman, '71) indi-

cate an increased sensitivity, relative to the first study, of up to 20 decibels, especially for the mid-range frequencies. Variability of behavior was also considerably lessened. Since differences between species were negligible in the first study, and because we feel the 20 dB difference between the first and second study is attributable to the response contingencies of the two methods, we would accept the data of the second study as being more accurate.

The present report (study 3) considers the high frequency sensitivity of *L. catta* from 8 kHz to 75 kHz; to complete the audiogram, we are also presenting the results of study 2, which extended down to a frequency of 100 Hz.

METHOD

The calibration procedures used in the present study differed from those previously employed; however, the training procedures, threshold testing techniques, and criteria were the same. Although a more complete description of these procedures appears in the earlier articles (especially study 2), the following brief comments are offered here.

Training procedures. The single-lever shock avoidance conditioning procedure of Clack and Herman ('63) contains two types of trials, blank and tone trials signaled by a white light above the response

bar. Tone trials consist of a ten second pairing of a pure tone of specified frequency and intensity with the white signal light. If the animal presses a bar within the ten seconds, both light and tone are terminated. Failure to press the bar during this period results in the presentation of a red signal light, concurrent with delivery of shock through the walls and floor of the cage. A blank trial consists of presentation of the white light alone (no tone). Any bar responses made during the blank trials or the intertrial intervals are designated as "false positives" and result in shorter shocks.

Blank trials and tone trials each occur with approximately equal frequency with a variable interval (8–15 sec) between trials. The net result of these stimulus and response conditions was to ensure that responses occur to sound alone and were not reflective of a generalized tendency to respond to any other cues.

Shock presentation was scrambled, with polarity changing three times a second. Shock values for all animals varied from .4 to .8 milliamperes, according to the animal and the level of performance.

Subjects. All of the animals used had good health records and otoscopic examination revealed that their ears were free of wax and appeared normal. Table 1 indicates age, sex, and weight of animals and the study in which they were used. Three of the four animals used in study 3 (3183, 4059 and 2311) were born at the Oregon Regional Primate Research Center. The remaining animals were acquired from Brown University in 1963, and their ages are estimated from dentition and, to a lesser degree, from body weight. It should be noted that three of the four animals used in the high frequency study (study 3) were also utilized in study 2.

TABLE 1

Animal	Age	Sex	Wt.	Used in study
	<i>years</i>		<i>kg</i>	<i>number</i>
3183	4	F	2.6	3
4059	3	F	2.2	2, 3
2311	5	M	2.6	2, 3
2023	6 (est)	M	3.0	2, 3
2024	6 (est)	M	2.8	2
647	7 (est)	M	3.2	2

Apparatus. The animals were trained and tested in a 13" × 13" × 13" cage housed inside a double-walled Industrial Acoustics Company sound booth lined with corrugated three and one-half inch foam rubber. The rubber was employed to further reduce external noise and to minimize standing wave patterns (areas of relatively high and low intensities generated by the interaction of reflected sound waves). The cage floor, walls, and ceiling were constructed of one-quarter inch brass rods spaced one-half inch apart. Shock was administered to the subjects by way of these conductive brass rods. The animals could be observed at all times through a one-way window.

Sound-producing equipment consisted of standard generators, filters, and attenuators. In addition, the high frequency electrostatic transducer used in study 3 (LTV Model EST-2) was used with a matched amplifier. This amplifier-transducer combination is capable of producing high intensity pure tone stimuli to over 200 kHz. The speaker was located eight inches from one side of the cage in a central position, and the lever and lights were located on the cage wall to the left, as one faced the transducer, and protruded into the cage. Stimuli presentation and response contingencies were controlled by solid state programming apparatus. An electronic switch was used to control possible on-off transients generated by the equipment. A variable filter was used behind the amplifier to both reduce amplifier noise and to eliminate unwanted lower frequency components that might have arisen in the system, although none were detectable by conventional means. Purity of signal was monitored by oscilloscope, and frequency was held to within three cycles of the nominal setting, as measured by a frequency counter.

Calibration. Calibration of the sound field was accomplished with a B & K one-quarter inch condenser microphone, cathode follower and a Sierra Model 301B wave analyzer. The microphone was originally placed in five or six different positions in the sound field, and the decibels of attenuation necessary to produce one dyne/cm² of sound pressure at that spot for a particular frequency were recorded. Sound

measurement procedures such as this, using different microphone placements, were performed weekly throughout the early course of the experiment.

Over the course of time, it became apparent that several different "strategies" were available to us in the measurement of the sound field. The measurement of high frequency (i.e., 25 kHz and above) signals presents certain problems that are not as apparent in the assessment of a low frequency sound field. Because the wavelength of high frequencies is relatively small, extensive differences in intensity may result from relatively minor differences in position. However, the variability associated with this technique was considerable for the high frequencies, occasionally being as high as 30 dB. A second alternative was derived from the fact that particular animals tended to "sample" the sound field in particular positions. Therefore, we considered developing separate calibrations for each animal according to its favorite sampling or listening sub-area. A third possibility arose from the finding that the sound field tended to demonstrate two "hot spots," or maxima, of approximately equal intensity. These were located on the transducer axis at about two inches and ten inches inside the cage and were teardrop in shape, and about $3'' \times 2'' \times 2''$ in volume. Within these areas, all frequencies demonstrated maximal intensity for any given setting. The loss in signal strength as one sampled outside these areas was approximately 25–30 dB for all frequencies. We decided to utilize these maximum values obtainable within these hot spots as our "true" intensity values, since two of the four animals consistently sampled from these sub-areas, and a third animal from a point immediately adjacent to one. Further, thresholds derived from these intensity values would be conservative, i.e., would tend to indicate a minimum sensitivity. The actual sensitivity of the lemur may therefore vary from the reported values by an unknown amount (but certainly less than 25–30 dB) at the higher frequencies (8 kHz and above). The difference, however, would always be in the direction of increasing sensitivity, i.e., the lemur is no less sensitive than reported here.

Threshold determinations. The threshold was defined as the intensity at which the probability of a response to a tone presentation was 50%. The shock was suspended during threshold determinations. A particular day's threshold estimate was based on at least 30 tone trials at each frequency, and final determinations were made by averaging at least four daily observations per animal. Thus, each threshold is based on a minimum of 120 trials per animal. If an animal's false positive rate (bar presses during blank trials) were greater than 20%, the threshold estimate associated with those data was rejected. The false positive rate was in most cases low enough to meet this criterion, and only in a few instances was it necessary to discard data.

RESULTS AND DISCUSSION

Figure 1 presents the mean auditory sensitivity function of the ringtail lemur over the frequency range 100 Hz to 75 kHz. The shaded areas indicated on the figure delineate the range (highest threshold–lowest threshold) of observations associated with each frequency. This figure is derived from data obtained in both study 2 (100 Hz to 40 kHz) and study 3 (8 kHz to 75 kHz).

There is considerable frequency range common to these studies. We feel justified both in averaging these mid-range data points and in presenting the results of the two studies together in a single function, since: (a) the same method was used in both studies, (b) three of the animals served as subjects in both studies, and (c) the degree of agreement between the two sets of data for these animals is remarkably good. Table 2 presents these mid-range frequency points obtained from those animals which served as subjects in both studies. As can be seen, with the exception of animal 2023, at 15 kHz, and animal 4059 at 40 kHz, there is good agreement between the sets of data. Accordingly, these values were averaged, and mean data plotted in audiogram form.

The auditory sensitivity function of figure 1 conforms quite well to previously reported data available for other members of the prosimian sub-order. These other data were gathered by Heffner, Ravizza,

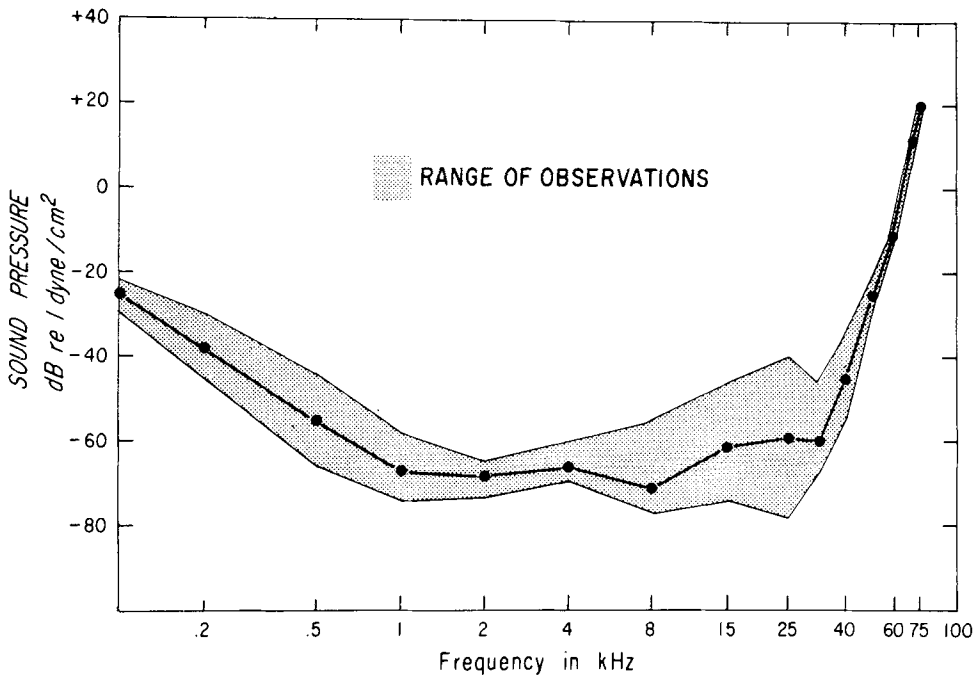


Fig. 1 Mean auditory sensitivity of the ringtail lemur derived from the results of study 2 (Mitchell et al., '71) and 3 (the present study) is represented by the black line. The shaded area indicates the total range of threshold observations at the particular frequencies examined: 0.1, 0.2, 0.5, 1, 2, 4, 8, 15, 25, 32, 40, 50, 60, 70, and 75 kHz.

TABLE 2
Comparisons of thresholds for the middle frequencies in both
study 2 (Mitchell et al., '71) and study 3 (the present work)

Frequency kHz	Lemurs					
	4059		2311		2023	
	Study 2	Study 3	Study 2	Study 3	Study 2	Study 3
8	-75	-79	-68	-77	-70	-74
15	-70	-66	-66	-61	-59	-34
25	-62	-62	-55	-52	-57	-53
32	-69	-61	-66	-62	-44	-49
40	-48	-60	-40	-43	-37	-33

All values in dB re: 1 dyne/cm².

and Masterton ('69) and Heffner and Masterton ('70), using a method of conditioned suppression, and are presented in figure 2, along with the lemur audiogram of the present study. In addition, figure 2 also presents auditory threshold curves for chimpanzee (Elder, '34; Farrer and Prim, '65) and man (Sivian and White, '33; Farrer and Prim, '65) which are included for purposes of contrast.

It would appear at first that the prosimians as a group possess rather unusual high frequency sensitivity, perhaps at the

expense of a loss in low frequency sensitivity. The lower frequency disparity between the anthropoids and the prosimians, essentially 8–15 dB from 100 Hz to 3 kHz, is readily apparent. Indeed, Mitchell ('70) has compared threshold values at 250 Hz, 500 Hz, and 1 kHz in a number of prosimians and anthropoids and has shown that they are different at or beyond the 0.002 level of significance (Mann-Whitney U test). However, the relatively low high-frequency cut-off found in the anthropoids is the exception when compared to the

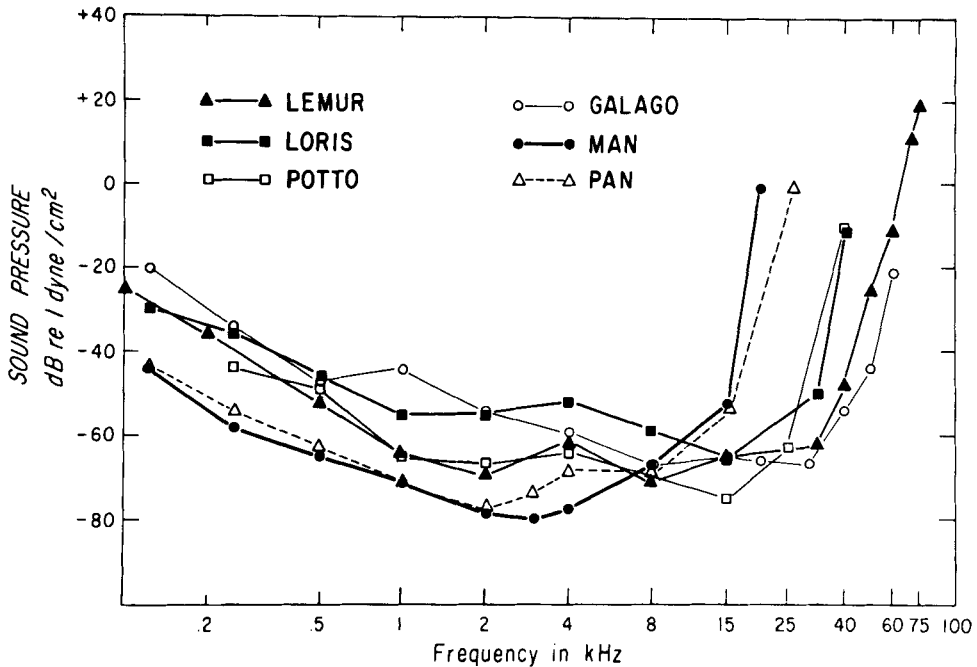


Fig. 2 The auditory sensitivity of several representatives from the sub-orders Prosimii and Anthropoidea. The prosimian audiograms are derived from the works of Masterton et al. ('69) Heffner et al. ('70) and the present study. The anthropoid data is from the studies of Elder ('34), Farrer and Prim ('65), and Sivian and White ('33).

mammalian class as a whole. Masterton, Heffner and Ravizza ('69) believe that high frequency sensitivity is a trait characteristic of ancestral mammals, and that it has been gradually lost in primate evolution. It is further conjectured that this sensitivity is probably necessitated more by the demand for accurate sound localization than by any other single source of selective pressure. They suggest that high frequency sensitivity in the higher primates was retained until its benefits for sound localization were replaced by those of wide-spaced ears, thus giving man and the rest of the anthropoids the advantages of larger disparities in time of arrival of sound waves at the two ears, and restoring sound localization ability. A question naturally follows then as to how the New World and Old World monkeys, apes, and man compare with the prosimians in regard to this anatomical feature? To our knowledge, no one has yet attempted the appropriate comparison to test this hypothesis.

It might be argued that the arbitrary selection of chimpanzee and man as representative of the sub-order Anthropoidea biases the results arising from contrast with the prosimians. However, none of several other possible choices (e.g., *Callithrix jacchus*, *Macaca mulatta*, *M. nemestrina*, *Saimiri sciureus*) differ significantly from man in terms of low frequency sensitivity (Stebbins, '71). Although differences do exist in reference to high frequency cut-off, none of the other anthropoids display the same degree of high frequency sensitivity as do the prosimians.

High frequency sensitivity would probably be of value in several classes of operations: (a) predator detection, such as possessed by noctuid moths in detection of echolocating bats (Roeder, '67); (b) vocalizations (Marler, '65), specifically in reference to such forms of social communications as alarm or distress cries, and (c) sound localization (of which echolocation would be a special case), as a means maximizing locomotion and/or food-getting.

Insofar as the lemur is concerned, we are unable to identify any significant predator in their natural environment, although hawks and a genet-like cat occasionally prey on them (Jolly, '66). The existence of an earlier land bridge from Madagascar to Africa opens the possibility of a significant past predator against whom high frequency sensitivity might have offered an appropriate warning system, but this is speculative.

In an effort to evaluate the frequency aspects of lemur vocalizations, we recorded three recognizably different calls on a tape recorder with an 80 kHz data band width and played the tape back through a Tektronix Spectrum Analyzer coupled to a storage oscilloscope. We were thus able to produce stored visual displays of the total frequency spectra generated by individual calls. These analyses indicate emission components ranging occasionally to 25–30 kHz, but being for the most part limited to 20 kHz and under. By far the bulk of the vocalizations produced their major energy around 10–15 kHz. This is also true of the lemur vocalizations examined by Andrew ('63). In the calls sampled by us, which did not include distress cries or true alarm cries, we found no evidence of the need for high frequency hearing. Although we have no reason to exclude the lemur as an echolocator, we do not suggest that he relies on this means of sensing as a primary locating technique. While *L. catta* is less arboreal than most lemurs, its essentially frugivorous diet and usual habitat seem to argue against the need for echolocating. Further, the fact that its most sensitive auditory range is at least one octave below the frequency range utilized by primary echolocators such as bats also seems to indicate that any use of echolocation would be limited.

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